

Linear Regression

Solved Numerical

Step 1: The Dataset

Xi meal bill and yi is tip

Meal	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	34	5	-40	-5	200
2	108	17	34	7	238
3	64	11	-10	1	-10
4	88	8	14	-2	-28
5	99	14	25	4	100
6	51	5	-23	-5	115

Here,

- **X (Independent Variable)** = Meal bill amount
- **Y (Dependent Variable)** = Tip amount

We want to predict:

Tip = $\theta_0 + \theta_1$ (Meal Bill)

where

- θ_0 = Intercept
- θ_1 = Slope

Step 2: Find the Best-Fit Line

The regression line is

$$Y = \theta_0 + \theta_1 X$$

The values of θ_0 and θ_1 should make the line fit the data as closely as possible.

Step 3: Compute the Slope (θ_1)

The image shows the formula:

$$\theta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

where

- x_i = Meal bill
- y_i = Tip
- $\bar{x}$ = Mean meal bill in this case the value is 74
- $\bar{y}$ = Mean tip in this case the value is 10

The table calculates:

	Meal	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	200		1600
2	238		1156
3	-10		100
4	-28		196
5	100		625
6	115		529

Now add the values.

Numerator

$$200 + 238 - 10 - 28 + 100 + 115 = 615$$

Denominator

$$1600 + 1156 + 100 + 196 + 625 + 529 = 4206$$

Therefore,

$$\theta_1 = \frac{615}{4206} = 0.146$$

This means:

For every \$1 increase in the meal bill, the predicted tip increases by about \$0.146.

Step 4: Compute the Intercept (θ_0)

Mean meal bill

$$\bar{x} = 74$$

Mean tip

$$\bar{y} = 10$$

Use

$$\begin{aligned}\theta_0 &= \bar{y} - \theta_1 \bar{x} \\ &= 10 - (0.146 \times 74) \\ &= -0.804\end{aligned}$$

So the regression equation becomes

$$\boxed{\text{Tip} = -0.804 + 0.146(\text{Meal Bill})}$$

Step 5: Predict the Tip

Suppose the meal bill is **\$80**.

$$\begin{aligned}\text{Tip} &= -0.804 + 0.146(80) \\ &= -0.804 + 11.68 \\ &= 10.88\end{aligned}$$

So the predicted tip is **\$10.88**.

Step 6: What is the Cost Function?

The **cost function** measures **how wrong the predictions are**.

The formula is:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum (\hat{y} - y)^2$$

where

- m = Number of meals
- y = Actual tip
- $\hat{y}$ = Predicted tip

Example

Suppose for a meal bill of **\$64**:

Actual Tip

$$y = 11$$

Predicted Tip

$$\hat{y} = 8$$

Error

$$11 - 8 = 3$$

Squared Error

$$3^2 = 9$$

The cost function adds the squared errors for **all meals** and computes their average (with the factor $1/(2m)$).

A **smaller cost** means the regression line fits the data better.

Why Square the Error?

Suppose

Actual Tip	Predicted Tip	Error
10	12	+2
10	8	-2

If we simply added the errors:

$$2 + (-2) = 0$$

It would incorrectly appear that there is **no error**.

By squaring them:

$$2^2 + (-2)^2 = 4 + 4 = 8$$

all errors become positive and larger mistakes are penalized more.

The objective of linear regression is to **find the values of θ_0 and θ_1 that minimize the cost function**,

resulting in the **best-fit line** for predicting tips from meal bills.

why the cost function is divided with 2 m what is m ?

The cost function for **Linear Regression** is:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

What is m?

m is the **total number of training examples (data points)**.

Example

Suppose your dataset contains:

Meal Bill Tip

34	5
108	17
64	11
88	8
99	14
51	5

There are **6 rows**.

Therefore,

$$m = 6$$

Why divide by 2m instead of just m?

The **2** is included **only to make the mathematics easier** when calculating gradients (derivatives) during gradient descent.

Suppose the cost function were

$$J = \frac{1}{m} \sum (\hat{y} - y)^2$$

When taking the derivative,

$$\frac{d}{dx}(x^2) = 2x$$

The derivative becomes

$$\frac{2}{m} \sum (\hat{y} - y)$$

Notice the extra **2**.

Now suppose we define

$$J = \frac{1}{2m} \sum (\hat{y} - y)^2$$

During differentiation,

$$\frac{d}{dx}(x^2) = 2x$$

The **2** from differentiation cancels the **2** in the denominator.

$$\frac{1}{2m} \times 2 = \frac{1}{m}$$

The gradient becomes much simpler:

$$\boxed{\frac{1}{m} \sum (\hat{y} - y)}$$